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Roll No

EX-6003-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Microprocessor and Microcontrollers

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What kind of technology is used in 8086 μ p? Explain the modes in which 8086 can operate?
b) Draw and discuss the architecture of 8086. Mention the jobs performed by EU and BIU.
2. a) Explain about programmable interval timer 8254.
b) What is memory mapped I/O and I/O mapped I/O and explain about interfacing 8255 to 8086 using memory mapped I/O and I/O mapped I/O.
3. a) Draw the register organization of 8086 and explain application of each register.
b) Explain the following instruction of 8086 with example.
XCHG, DAA, XOR, PUSH, POP, RCR, AAA, INRET, OUT
4. a) What is the difference between 8254 and 8255 programmable interval timer? List the various modes of 8255. Explain any one mode with waveforms.
b) Draw the architecture of DMA controller. Explain in brief.

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5. a) Explain the operation of stepper motor interfacing to 8051.
b) What is RS-232 and explain the 8051 connections to RS-232.
6. a) Define External and Internal memory of microcontroller.
b) Describe different types of Addressing modes.
7. a) With the help of example explain programming of external interrupts.
b) Explain the memory structure of 8051.

OR

- a) Why is PIC program memory 14 bits wide?
 - b) What do you mean by orthogonal instruction set?
8. Write a short notes on any two:
- a) Programmable counter array (PCA)
 - b) Features and limitation of 32 bit PIC microcontroller
 - c) 8096 instruction set

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